# Genuine Multipartite Entanglement Analysis — Molecular System

## MOLECULE

Benzene  (Benzene)
Basis set    : external-CI
Charge       : 0
Multiplicity : 1
Total e-     : 6
n-body order : 6

## GROUND STATE ENERGIES

E0 (Ha)  : nan
E1 (Ha)  : nan
Gap (Ha) : nan
HF  (Ha) : nan

## CONVEX-ROOF METHODS

Methods run    : gcg
Restarts       : 20

C_LB  : coherence witness bound
C_gcg: Conjugate Gradient(Alg. A)
C_UB  : eigendecomposition bound

## ACTIVE SPACE  CAS(6,6)

Frozen core orbs : 0
Frozen e-        : 0
Active e-        : 6
Active orbs      : 6
Spin-orbitals    : 12
Hilbert dim      : 400

## SPIN PROPERTIES

<S2>         : 0.000000
S            : 0.0000
<Sz>         : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

## GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

## DOUBLE OCCUPANCY

<D> total  : 1.058749
<D> / site : 0.176458

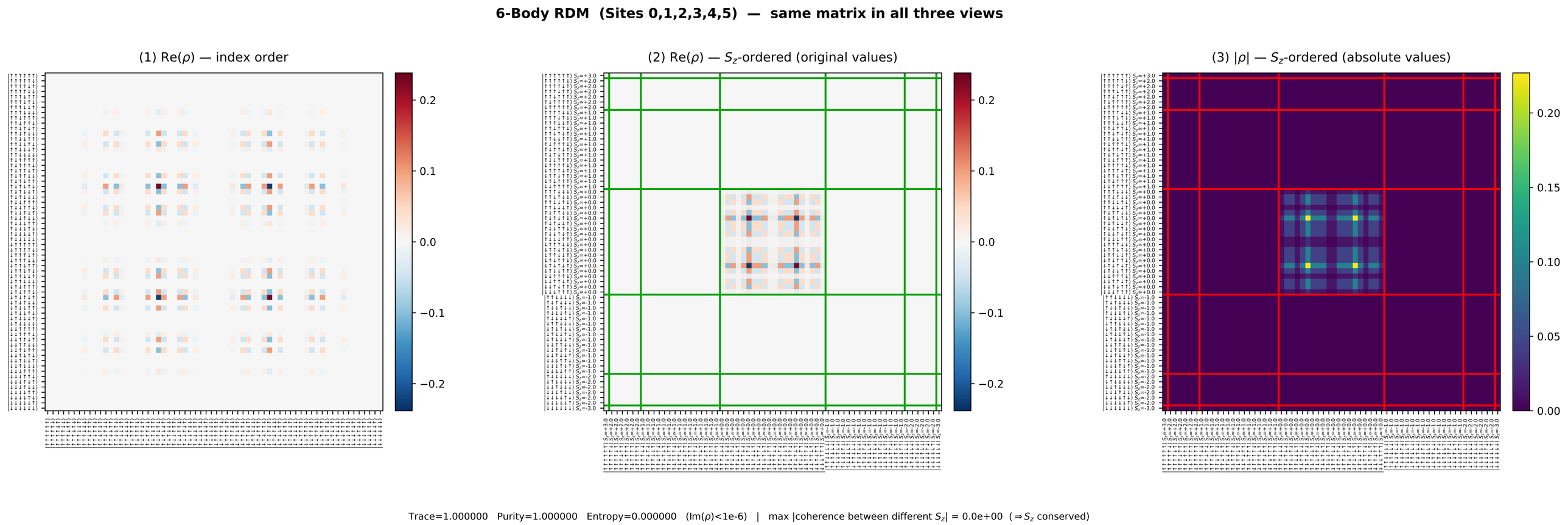
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Body order : 6
Clusters   : 1

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## Summary — 6-Body RDMs  (Benzene, CAS(6,6))

| Orbitals | Purity | Entropy | C_GME (pure) | C_LB | C_gcg | C_UB | State |
|---|---|---|---|---|---|---|---|
| 0,1,2,3,4,5 | 1.000000 | 0.000000 | — | — | — | — | PURE |


6-Body RDM (Sites 0,1,2,3,4,5) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — $S_z$-ordered (original values)
(3) |ρ| — $S_z$-ordered (absolute values)
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Im(ρ)<1e-6) | max |coherence between different $S_z$| = 0.0e+00 (⇒ $S_z$ conserved)